

Pronunciation Decision Infrastructure

Technical Architecture and Clinical Safety Boundary

advX Speech Service

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Executive summary

This document explains how the project’s decision infrastructure is built, from raw audio acquisition to clinician-reviewed evidence.

The current system is an experimental English pronunciation assistant. Its intended role is to help speech-language pathologists collect repeatable exercises, inspect acoustic evidence, and record their own clinical review. It is **not** a diagnostic system and does not autonomously prescribe therapy.

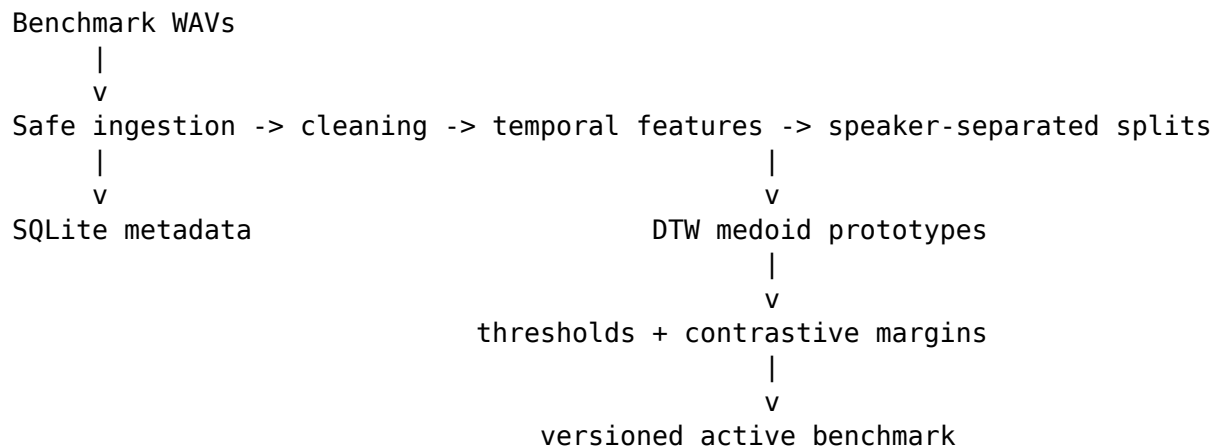
The architecture deliberately separates three responsibilities:

1. **Signal and acquisition decisions** determine whether audio is usable.
2. **Acoustic model decisions** compare speech with calibrated word benchmarks and expose uncertainty.
3. **Clinical decisions** belong to a qualified human reviewer.

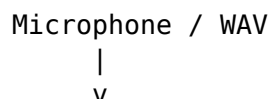
For isolated-word experiments, the model can return CORRECT, INCORRECT, UNDECIDABLE, or RETRY. For natural phrases and pediatric workflows, the system stores only REVIEW_REQUIRED or RETRY. This distinction prevents a word-level adult benchmark from being presented as a validated pediatric clinical judgment.

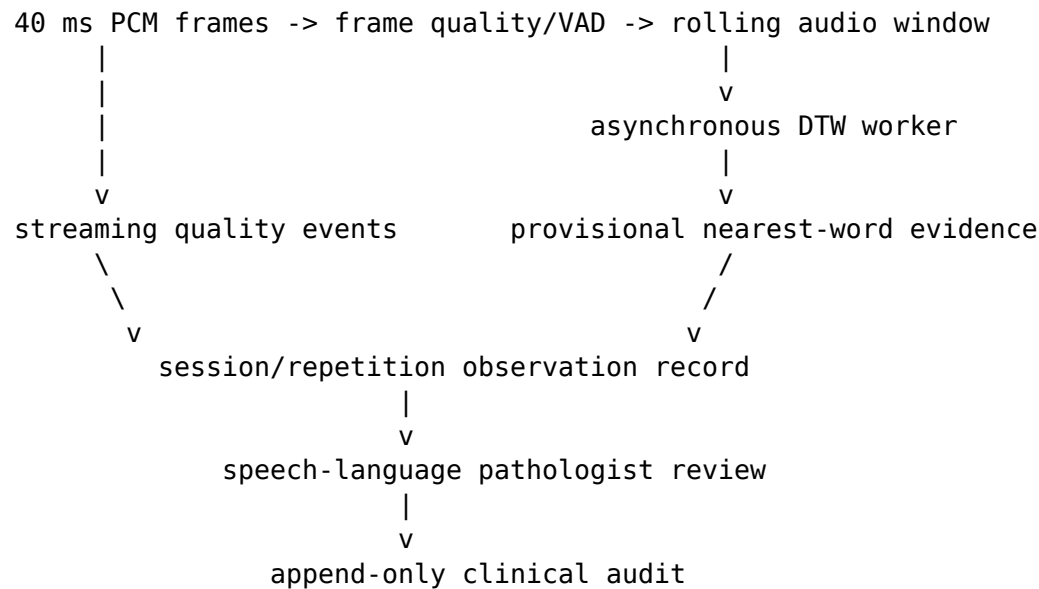
Architectural overview

OFFLINE CALIBRATION



ONLINE EXECUTION





The offline and online paths are intentionally different. Calibration creates stable, versioned decision artifacts. Online execution consumes those artifacts but never silently recalibrates them using a patient recording.

Core data boundaries

Audio contract

The canonical public audio representation is:

- sample rate: 16,000 Hz;
- channels: one;
- encoding: signed 16-bit PCM, little-endian;
- frame duration: 40 ms;
- samples per frame: 640;
- bytes per frame: 1,280.

Every streaming frame is validated against this exact size. A malformed frame is rejected rather than padded or truncated because implicit repair would alter timing and make later evidence difficult to audit.

Feature contracts

The system currently supports two acoustic representations.

Representation	Shape	Purpose
Summary MFCC embedding	26 values	Fast baseline and vector retrieval
Temporal MFCC sequence	frames × 36	DTW pronunciation comparison

The temporal representation uses 12 MFCC coefficients, excluding coefficient zero, plus first- and second-order temporal derivatives. Utterance-level normalization reduces broad recording-level

scale differences. The feature and distance implementations carry explicit version identifiers so that results from incompatible pipelines are not silently mixed.

Offline benchmark construction

Safe dataset ingestion

Archive members are validated before extraction to prevent path traversal. Each source file receives a content hash, and speaker identifiers are pseudonymized. Duplicate content is ignored idempotently.

The persistent recording row contains the word, locale, pseudonymous speaker, take number, dataset split, duration, cleaning status, feature version, and source hash. The original speaker identity is not needed for scoring.

Speaker-separated splits

Training, validation, and test splits are assigned by speaker rather than by recording. If recordings from one speaker appeared in both training and test, the reported accuracy could measure speaker recognition instead of pronunciation generalization.

The three splits have separate roles:

- **training** selects representative prototypes;
- **validation** calibrates thresholds and margins;
- **test** estimates performance after calibration.

The test split must never be used to choose thresholds.

Cleaning gate

The cleaning pipeline performs:

- anti-aliased resampling where necessary;
- a 70 Hz high-pass filter;
- energy-based voice activity detection;
- morphological removal of short impulses and filling of short gaps;
- conditional spectral subtraction at low estimated SNR;
- level normalization near -23 dBFS.

Cleaning produces both canonical PCM and a quality report. If the signal is too short, silent, severely clipped, or otherwise unreliable, the decision path returns RETRY. A quality failure is not interpreted as a pronunciation error.

DTW prototypes

For each word, the temporal pipeline selects several real training utterances as prototypes. It does not average MFCC sequences into an artificial voice.

Prototype selection uses deterministic medoid candidates and a refinement step:

1. calculate pairwise DTW distances among candidates;
2. select a central candidate;
3. add distant candidates to cover variation;
4. assign candidates to their closest selected prototype;

5. replace each prototype with the most central real recording in its group.

Multiple prototypes are important because pronunciation, pitch, speaking rate, and vocal-tract characteristics vary substantially across speakers.

Distance function

For an input temporal sequence X and a prototype P , dynamic time warping finds a low-cost monotonic alignment through their frame-distance matrix. Frame cost is cosine distance. A band limits excessive warping and reduces computation.

The distance from an input to a word is:

$$d(X, w) = \min_{p \in P_w} DTW(X, p)$$

where P_w is the set of real prototypes for word w . Lower distance means greater acoustic similarity; it does not directly mean clinical correctness.

Calibration of the decision regions

Each active word benchmark stores three calibrated values:

- `accept_threshold`;
- `reject_threshold`;
- `margin_threshold`.

Positive validation distances are obtained from recordings of the target word. Negative distances are obtained from other words. The current prototype uses conservative quantiles:

$$T_{accept} = \min(P90(d_{positive}), P10(d_{negative}))$$

$$T_{reject} = \max(T_{accept} + 0.02, P97(d_{positive}))$$

The contrastive margin compares the target with the closest competing word:

$$margin = d(X, competitor) - d(X, target)$$

A positive margin means that the target is closer. The margin threshold is calibrated from the lower tail of positive validation margins, with a minimum safety floor.

These formulas are an experimental calibration policy, not permanent product constants. A pediatric dataset and an explicit clinical cost model are needed before selecting production thresholds.

Isolated-word decision state machine

For an isolated target word, the system computes its distance and the nearest competitor.

Condition	State	Meaning
Audio quality gate fails	RETRY	No pronunciation judgment
Missing or invalid evidence	UNDECIDABLE	The model cannot decide
Target distance $\leq$ accept threshold and margin $\geq$ margin threshold	CORRECT	Accepted by experimental benchmark
Target distance $\geq$ reject threshold	INCORRECT	Outside target benchmark
Competitor is closer by the required margin	INCORRECT	Contrastive mismatch
All other valid cases	UNDECIDABLE	Evidence falls in the uncertainty region

The uncertainty region is intentional. Collapsing it into a binary result would increase either false acceptance or false rejection, and would hide known limitations of the dataset.

Every result includes reason codes, target distance, competitor identity, competitor distance, margin, model version, and benchmark version. A user interface should show a simple message to a child while retaining the detailed evidence for the clinician and audit log.

Streaming decision infrastructure

Why acquisition and model inference are decoupled

The microphone produces one frame every 40 ms. The full DTW comparison can take longer than this budget, especially when many prototypes are active. Running it synchronously would block microphone reads and cause frame loss.

The streaming engine therefore has two inference rates:

1. **Every 40 ms:** RMS, peak, clipping, and lightweight voice activity.
2. **Every 200 ms when sufficient voiced audio exists:** schedule a temporal comparison on the most recent rolling window.

The DTW executor has one worker and at most one active job. If the worker is busy, the engine does not queue obsolete windows. Each frame event reports the most recently completed acoustic evidence and whether a refresh was scheduled or completed.

On the local 51-frame control recording, `push_frame` took 0.20 ms on average and 0.75 ms at maximum after asynchronous decoupling. This is below the 40 ms acquisition budget, but it must be remeasured on deployment hardware and with the final model.

Streaming event protocol

The WebSocket endpoint is:

`/streaming/sessions/{session_id}`

The client begins with:

```
{
  "type": "session.start",
  "target_words": ["bird", "happy"],
```

```
"seed": "demo"  
}
```

The server returns a natural or controlled English prompt and the audio contract. The client then sends binary 1,280-byte frames. Each accepted frame produces a `stream.inference.partial` event.

The client finishes with:

```
{"type": "session.finish"}
```

The final event includes duration, voiced-frame ratio, last rolling distances, model errors if any, and a safe status.

Phrase-level safety rule

The available benchmarks contain isolated words, while a natural phrase has coarticulation and no guaranteed pauses between words. A rolling nearest-word result cannot identify reliable word or phoneme boundaries.

Consequently:

- partial nearest-word evidence is marked provisional;
- `clinical_use` is always false;
- `alignment_status` is `not_available`;
- a usable phrase ends in `REVIEW_REQUIRED`;
- insufficient or mostly silent audio ends in `RETRY`.

Reliable phrase feedback requires a forced aligner validated on child speech, phone-level targets, and evaluation on the expected disorders and age bands.

Clinical workflow and human authority

The clinical persistence layer models:

- a pseudonymous child profile and age band;
- guardian consent state;
- a clinician-created exercise;
- sound, syllable, word, or phrase level;
- target text and optional target phonemes/error patterns;
- a session and its model version;
- multiple repetitions;
- automatic observations;
- clinician reviews;
- append-only audit events.

The automatic system is technically prevented from writing `CORRECT` or `INCORRECT` into the pediatric repetition table. It may write only:

- `REVIEW_REQUIRED`;
- `RETRY`.

The clinician separately records:

- `accepted`;
- `speech_error`;
- `retry`;

- structured observations and optional notes.

This separation preserves who made each judgment and supports later recalibration without rewriting clinical history.

Persistence architecture

SQLite: authoritative state

SQLite stores durable, relational information:

- dataset metadata and split assignments;
- processed recording identities;
- temporal features and versioned benchmark artifacts;
- calibrated thresholds;
- attempts and reason codes;
- child consent and exercises;
- sessions, repetitions, reviews, and audit events.

SQLite is appropriate for the current local single-service deployment. WAL mode, foreign keys, short-lived connections, busy timeouts, unique constraints, and idempotent ingestion protect basic consistency.

Redis 8: transport, cache, and optional vector retrieval

Redis 8 is used for transient workloads:

- session event streams;
- benchmark cache;
- transport between producers and consumers;
- optional HNSW nearest-neighbor retrieval.

The v3 runtime is pinned to the official `redis:8.8.0-alpine` image. Redis 8 unifies the former Stack capabilities with the core distribution, including Redis Search and vector indexing. It also exposes native Vector Set commands. The deployment chooses Redis’s AGPLv3 open-source license option; distribution and network-service obligations must be reviewed for the final product.

At startup or during diagnostics, `scripts.check_redis` performs read-only capability discovery. It distinguishes an unreachable service, a reachable core-only server, and a complete Redis 8 runtime exposing `FT.SEARCH` and `VSIM`. Failure of optional vector capabilities does not corrupt SQLite state.

`scripts.sync_vector_index` reads accepted 26-dimensional MFCC summaries from SQLite, computes a locale-wide mean and standard deviation, normalizes every vector consistently, and upserts it by immutable recording ID. Repeating the sync updates the same keys. The normalization parameters and version are stored with the index so query vectors use the identical transform.

Summary embeddings are indexed as 26-dimensional `FL0AT32` vectors with cosine distance. The vector index accelerates candidate retrieval; it does not replace SQLite, threshold calibration, temporal DTW, or clinical review.

SQLite	Redis 8
-----	-----
source of truth	transient acceleration
threshold versions	event streams
clinical audit	cache

consent and review vector candidate search
transactional relationships TTL/session state

Bias and pediatric limitations

The present benchmark fragment is not clinically representative. Known risks include:

- adult rather than pediatric voices;
- demographic and gender imbalance;
- limited vocabulary;
- no phoneme-level error labels;
- no coverage of speech sound disorders;
- microphone and environment mismatch;
- possible confounding by pitch, accent, or speaking rate.

Observed local failures already demonstrate why uncertainty and human review are required: a correctly pronounced word by a female speaker can be rejected, and an acoustically similar non-target pronunciation can be accepted or remain undecidable.

A vector database does not remove these biases. The correction requires representative data, speaker-level splits, subgroup metrics, suitable acoustic representations, calibrated uncertainty, and clinical validation.

Privacy, security, and retention

Child voice recordings are sensitive personal data. The current clinical row stores an SHA-256 digest, duration, and observations rather than raw PCM. If audio must be retained, it should live in a separate encrypted object store with:

- explicit purpose and legal basis;
- guardian consent and withdrawal workflow;
- role-based access control;
- encryption in transit and at rest;
- retention expiration and verified deletion;
- access and export audit;
- backup deletion policy;
- data protection impact assessment.

Production WebSockets also require authentication, authorization to the specific session, TLS, origin controls, rate limits, payload limits, and protection against replay and session identifier guessing.

Failure modes and safe behavior

Failure	Safe behavior
Wrong frame size	Reject frame and report protocol error
Silence or very short recording	RETRY
Cleaning cannot produce reliable speech	RETRY
Target benchmark missing	Explicit error; no fallback judgment
Model job exceeds frame interval	Continue acquisition; keep last evidence

Failure	Safe behavior
Model worker throws an exception	Report <code>model_error</code> ; do not invent result
Distances fall between thresholds	UNDECIDABLE
Phrase has no validated alignment	REVIEW_REQUIRED
Consent is absent or withdrawn	Do not create exercise/session
Clinician has not reviewed	Preserve automatic observation only
Redis unavailable	Durable SQLite data remains authoritative

Verification strategy

The repository currently has 74 passing tests. Relevant decision tests cover:

- cleaning contract and quality behavior;
- SQLite idempotency and transactions;
- Redis/Valkey protocol doubles and fuzzing;
- acoustic repository constraints;
- temporal feature and DTW behavior;
- contrastive decision logic;
- exact 40 ms streaming frames;
- non-blocking streaming refresh behavior;
- safe phrase final state;
- consent-gated clinical persistence;
- separation between system and clinician verdicts;
- append-only audit records;
- vector index command and shape validation.

Before clinical use, software tests must be complemented by a locked external evaluation dataset and prospective human validation.

Recommended evolution

Phase 1 — engineering prototype

- Keep word-level DTW and explicit uncertainty.
- Add frame-loss, jitter, and long-session stress tests.
- Add authentication and encrypted audio storage only if retention is needed.
- Instrument end-to-end latency and model queue saturation.

Phase 2 — speech analysis

- Define a phoneme inventory and clinician-approved error taxonomy.
- Integrate an open forced aligner suitable for English.
- Store aligned evidence with model and alignment versions.
- Evaluate phone-level observations without presenting diagnoses.

Phase 3 — pediatric calibration

- Acquire legally usable pediatric and disordered-speech corpora.
- Split strictly by child.

- Report sensitivity, specificity, false rejection, false acceptance, and calibration by age band, gender, accent, microphone, and error type.
- Define operating thresholds with clinicians and document the cost of each error.

Phase 4 — supervised product trial

- Complete privacy and regulatory assessment.
- Freeze model and benchmark versions for the trial.
- Run usability and safety studies with speech-language pathologists.
- Measure whether the tool improves workflow without replacing professional judgment.

Source map

The principal implementation files are:

Responsibility	File
Audio cleaning adapter	app/cleaning.py
Signal processing	app/audio_cleaning.py
Temporal features and DTW	app/temporal_features.py
Prototype calibration and decisions	app/temporal_benchmark.py
Temporal persistence	app/temporal_repository.py
40 ms streaming engine	app/streaming_inference.py
WebSocket protocol	app/api.py
Prompt generation	app/sentence_composer.py
Pediatric clinical persistence	app/clinical_repository.py
Redis 8 vector index	app/vector_index.py
SQLite-to-HNSW synchronization	app/vector_sync.py
SQLite schema	scripts/init_db.py
Microphone demonstration	scripts/streaming_demo.py

Conclusion

The decision infrastructure is designed around traceability and uncertainty. It combines deterministic signal checks, versioned acoustic evidence, contrastive DTW decisions, non-blocking streaming, relational audit, and an explicit human clinical boundary.

The most important next improvement is not a more aggressive threshold or a different database. It is representative pediatric, phone-labelled data and a validated alignment/evaluation protocol designed with speech-language pathologists. Until then, the architecture can support controlled experiments and clinician review, but should not make autonomous clinical claims.